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FOOD DELIVERY AGENT

1) Problem Formulation:-

Deliver a food order from a restaurant to the customer using a suitable route.

Initial State:

Delivery rider is at the restaurant with the food order.

Actions:

- 1) Pick up food
- 2) Move to connected location
- 3) Continue on route.
- 4) Deliver food.

Transition Model

After Action, the agent moves from its current state to a new state.

Restaurant $\rightarrow$ Move $\rightarrow$ A $\rightarrow$ B $\rightarrow$ Deliver $\rightarrow$ Customer

Goal Test:

Check whether the rider has reached the Customer & food is delivered or not?

Path Cost:-

States / Actions.

The distance / number of roads travelled from restaurant to customer.

Search: BFS can be used when all roads have equal cost.

2) State TYPE:

"Single state type"

Food Delivery agent is a single state problem in this scenario because:

- Current location is known
- Restaurant & Customer's locations are known
- Environment is known.
- Actions have predictable results.

3) Environmental Properties

Property	Food Delivery Agent
- Observability	Fully observable
- Determinism	Deterministic
- Environment	Known
- State	Single state
- Changes	Static
- Actions	Predictable

4) Search Strategy:

Uniformed Search

• BFS

• DFS

• UCS

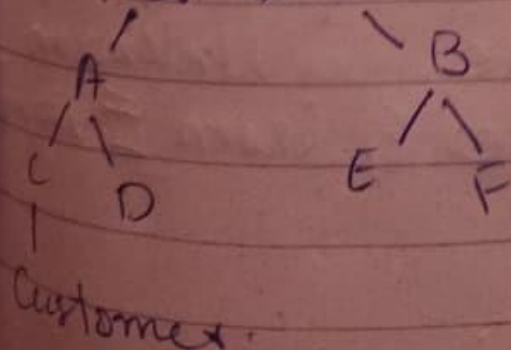
↳ Uniformed search doesn't use a heuristic, while Informed search uses as a heuristic to guide the search

5) BFS

Restaurant

Start: Restaurant

Goal: Customer



Queue:

[Restaurant]

[A, B]

[B, C, D]

[C, D, E, F]

Customer is found when C is expanded

~~Shortest Path:~~

Expansions Order } Restaurant $\rightarrow A \rightarrow B \rightarrow C$ }
Customer

Shortest Path:

Restaurant $\rightarrow A \rightarrow C \rightarrow$ Customer.

7) Alternative State

Multiple State:

If the rider doesn't know the exact current location, possible states can be represented as a belief state

Contingency:

If a road is blocked the agent uses an alternative route based on the situation.

Exploration:-

If the Delivery area is Unknown the agent explores & learn Environment